

Time allowed: 2 hours

Calculators are allowed

1.

a) Apply De Morgan's Theorems to the following expressions and **simplify**: (10 marks)

i.
$$Z = \overline{\overline{(P + \overline{Q + R})} + (\overline{P} \cdot \overline{Q} + \overline{P} \cdot \overline{Q})}$$

ii.
$$A = \overline{(X + Y)} + \overline{(\overline{X} \cdot \overline{Y} + Z)}$$

b) $Z = F(A, B, C, D)$ $Z = 1$ for the minterms (1, 2, 3, 8, 9, 11, 12, 14, 15) Z = don't care for the minterms (4, 7, 10) $Z = 0$ for the remaining minterms

(i). Simplify using K-maps (5 marks)

(ii). Simplify Z using Tabular Method. (10 marks)

(25 marks)

2.

a) Implement a Full Adder circuit using 2 half adders and an OR gate (5 marks)

b) A train has 2 emergency sensors ES1 and ES2 which can be operated by either the driver of the train and any passenger. There are 2 more sensors S1 and S2 mounted onto the engine. If the following conditions occur, then the train has to brake.

- If sensors ES1 is activated
- If sensors ES2 is activated
- If sensors S1 and S2 are activated at the same time (10 marks)

i. Determine the truth table for the system

ii. Obtain the simplified POS expression

iii. Implement the circuit using:

a. **NOR** gates only

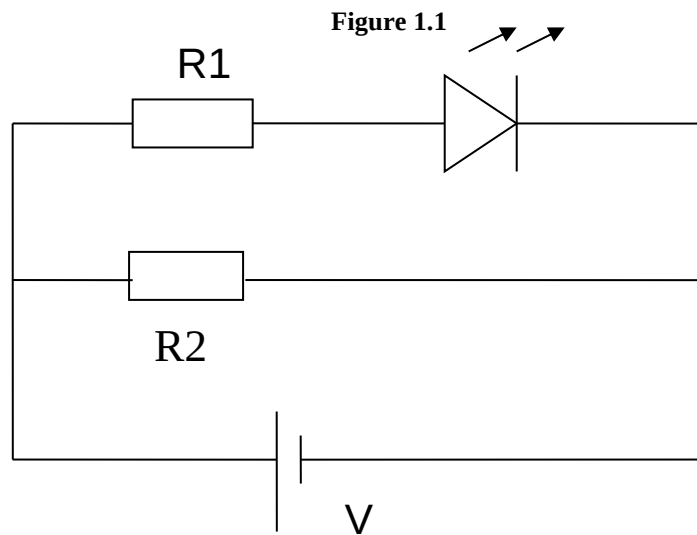
b. A suitable Multiplexer.

c) Design a Sequential Logic Circuit to generate the count sequence of 1,2,6,7 using positive edge triggered JK flip-flops. (10 marks)

(25 marks)

3.

- a) Describe the construction and characteristics of two types of extrinsic semiconductors. (5 marks)
- b) Draw a full-wave rectifier circuit with capacitor smoothing and explain its operation using appropriate waveforms. (10 marks)
- c) A student plans to measure the current going through the Light Emitting Diode (LED) which presents in the circuit given by Figure 1.1. Copy the circuit in Figure 1.1 to your answer script and answer the following: (10 marks)



- i. What is the instrument used to measure the expected measurement?
- ii. Explain briefly the method used to obtain the measurement and complete the circuit diagram including the instrument that you suggest.
- iii. If the value obtained in the experiment is 12mA, and if $R1 = 3 \text{ Kilo}$ find the external voltage (Assume there is no voltage drop across the LED)
- iv. If $R2$ is 5 Kilo Ohms find the total current coming out of the Battery.

(25 marks)

4.

- a) Draw the transfer characteristic curves of a Bipolar Junction Transistor. Clearly label Active region, Cut-off region and Saturation region in your diagram. (5 marks)
- b) Explain the operation of a JFET with a suitable diagram when the $V_{GS} < 0$ and $V_{DS} > 0$. (10 marks)

- c) Consider the following circuit assume it to be in the active mode. Hence calculate I_B , I_C and V_{CE} . Transistor is made from Si. (10 marks)

Material	$V_{CE(sat)}$	$V_{BE(sat)}$	$V_{BE(act)}$	$V_{BE(cut\ in)}$
Si	0.2V	0.8V	0.7V	0.5V
Ge	0.1V	0.3V	0.2V	0.1V

β	R_B	R_C	V_{CC}	V_{BB}
20	80K Ω	2K Ω	10V	5V

